```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).
```

```
import tensorflow as tf
import numpy as np
import os
import pandas as pd
from keras.applications.vgg16 import VGG16,preprocess_input


from keras.preprocessing import image

test_dir = '/content/drive/MyDrive/test'

test = []
for file in os.listdir(test_dir):
    test.append(['test/{}'.format(file), file])
test = pd.DataFrame(test, columns=['filepath', 'file'])

test.shape
```

```
(794, 2)
```

```
test.head(2)
```

```
             filepath           file
0  test/003d61042.png  003d61042.png
1  test/02cfeb38d.png  02cfeb38d.png
```

```
INPUT_SIZE = 224

def read_img(filepath, size):
    img = image.load_img(filepath, target_size=size)
    img = image.img_to_array(img)
    return img


x_test = np.zeros((len(test), 224,224, 3), dtype='float32')
i = 0
for file in test['file']:
    img = read_img(os.path.join(test_dir,file), (INPUT_SIZE,
INPUT_SIZE))
    x = preprocess_input(np.expand_dims(img.copy(), axis=0))
    x_test[i] = x
    i+=1
print('test Images shape: {} size: {:,}'.format(x_test.shape,
x_test.size))
```

```
test Images shape: (794, 224, 224, 3) size: 119,519,232
```

```
class_labels = ['Black-grass', 'Charlock', 'Cleavers', 'Common
Chickweed', 'Common wheat', 'Fat Hen', 'Loose Silky-bent',
              'Maize', 'Scentless Mayweed', 'Shepherds Purse', 'Small-
flowered Cranesbill', 'Sugar beet']
```

VGG

```
vgg = tf.keras.models.load_model('/content/drive/MyDrive/vgg.h5')

vggpred = vgg.predict(x_test)
print('Test bottleneck features shape: {} size:
{:,}'.format(vggpred.shape, vggpred.size))
```

```
Test bottleneck features shape: (794, 12) size: 9,528
```

```
vggpred = np.argmax(vggpred,axis=1)

for i in range(20):
  print(class_labels[vggpred[i]])
```

```
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Common Chickweed
Maize
Common Chickweed
```

ResNet

```
resnet =
tf.keras.models.load_model('/content/drive/MyDrive/resnet.h5')

resnetpred = resnet.predict(x_test)
print('Test bottleneck features shape: {} size:
{:,}'.format(resnetpred.shape, resnetpred.size))
```

```
Test bottleneck features shape: (794, 12) size: 9,528
```

```
resnetpred = np.argmax(resnetpred,axis=1)


for i in range(20):
  print(class_labels[resnetpred[i]])
```

```
Loose Silky-bent
Loose Silky-bent
Loose Silky-bent
Loose Silky-bent
Loose Silky-bent
Loose Silky-bent
Loose Silky-bent
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Loose Silky-bent
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Loose Silky-bent
Loose Silky-bent
```

Inception

```
inc = tf.keras.models.load_model('/content/drive/MyDrive/inc.h5')

incpred = inc.predict(x_test)
print('Test bottleneck features shape: {} size:
{:,}'.format(incpred.shape, incpred.size))


Test bottleneck features shape: (794, 12) size: 9,528

incpred = np.argmax(incpred,axis=1)

for i in range(20):
  print(class_labels[incpred[i]])
```

```
Charlock
Charlock
Charlock
Charlock
Charlock
Charlock
Charlock
Small-flowered Cranesbill
Charlock
```

Charlock
Charlock
Small-flowered Cranesbill
Charlock
Charlock
Charlock
Charlock
Charlock
Charlock
Charlock
Charlock